

CSE303 Final Group Project : Final Instructions

The initial project submission required you to select a dataset and perform **dataset exploration and initial statistical analysis**, including descriptive statistics, distribution analysis, categorical analysis, relationships between variables, data quality assessment, and key insights.

For the **final submission**, you will extend your previous work by applying predictive models and evaluating their performance.

1. Linear Regression

Apply **Linear Regression** to perform a regression task using an appropriate numerical feature from your chosen dataset. You should:

- Clearly identify the **target variable** and the feature(s) used for prediction. Explain why the selected variable is appropriate for a regression task.
- Split the dataset into **training and testing sets** using either **80% training - 20% testing**, or **70% training - 30% testing**. The split may be random. However, you are encouraged to use a suitable **sampling strategy**.
- Train the Linear Regression model using the training data. Make predictions on the **test dataset**.
- Calculate the following evaluation metrics on the **test dataset**:
 - **MSE (Mean Squared Error)**
 - **RMSE (Root Mean Squared Error)**

2. Logistic Regression

Apply **Logistic Regression** to perform a classification task using an appropriate feature from your chosen dataset. You should:

- Clearly identify the **target variable** and the feature(s) used for classification. Explain why the selected variable is appropriate for a classification task.
- Use the same type of train-test split: **80% training - 20% testing**, or **70% training - 30% testing**.
- Train the Logistic Regression model using the training data. Make predictions on the **test dataset**.
- Calculate the following evaluation metrics on the **test dataset**:
 - **Accuracy**
 - **F1 Score**

3. Additional Models (Optional)

You may apply **additional regression or classification models** if you want to explore your dataset further. However, **Linear Regression and Logistic Regression are mandatory**. If additional models are used, clearly explain why they were selected and compare their results with the required models.

4. Results and Discussion

Do not only report the evaluation metrics. **Discuss and interpret your results.** For each model, explain:

- What the obtained results indicate about the model's performance.
- Whether the result is good or poor and **why**.
- Possible reasons for the obtained performance.
- How the characteristics of your dataset may have affected the results.
- Any limitations or issues that may have affected the model.

For example, if the classification model achieves a low F1 score, discuss possible reasons such as class imbalance, insufficiently informative features, overlapping classes, or other characteristics of the dataset.

5. Final Report

Your final report should contain **both parts of the project**:

Part 1: Previous Work - Dataset Description, Central Tendency and Dispersion, Distribution and Shape, Categorical Variable Analysis, Relationships Between Variables, Data Quality Assessment, and Key Insights.

Part 2: Predictive Modeling - Train-test splitting methodology, Linear Regression, Logistic Regression, Results and discussion and Conclusions and key findings.

You should **combine the previous report and the new modeling section into one complete final report**.

6. Final Submission

You must submit **both**:

A. Final Report

Submit **one PDF file** containing the complete project report (Previous Part + New Modeling Part).

B. Full Code Notebook

Submit the **complete Jupyter Notebook (.ipynb)** containing all code used for the project, including: Data loading, Data preprocessing/cleaning, Exploratory data analysis, Visualizations, Statistical analysis, Linear Regression, Logistic Regression, Model Evaluation, Any additional models, if used.

C. The dataset used.

The notebook should be complete and executable so that the analysis and results can be followed from beginning to end.

Important: The final submission is not only the new modeling part. You must submit the **complete project (Previous Part + New Part)** in the final report and the **complete code + dataset** in the notebook submission.